Building A Trading Simulator To Replicate Vanilla Option Payoff

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Outline

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Motivation & Research Questions

Core Idea

Can we *replicate* a short European call position using only the underlying asset and a risk-free bond?

Research Questions

1. How accurately does a *discrete* delta hedge approximate continuous Black-Scholes replication in practice?
2. How does *rebalancing frequency* affect hedge quality (daily vs. weekly vs. monthly)?
3. Does *Monte Carlo delta estimation* converge to the closed-form Black-Scholes result as the path count grows?

Contd.

Contract Under Study

Underlying	SPY (S&P 500 ETF)
Option type	European Call, Strike \$600
Expiry	18 December 2026
Implied vol σ	24.49% (Bloomberg, 5 Oct 2025)
Risk-free rate r	4.5% p.a.
Notional	100 shares per contract
Backtest window	5 Oct 2025 – 7 Apr 2026 (184 days)

GBM and Itô's Lemma

Asset price model — Geometric Brownian Motion:

$$dS_t = \mu S_t \, dt + \sigma S_t \, dW_t$$

Itô's Lemma (chain rule for stochastic processes): if $f \in C^{1,2}$ and $dX_t = \mu_t \, dt + \sigma_t \, dW_t$, then

$$df(t, X_t) = \left(\frac{\partial f}{\partial t} + \mu_t \frac{\partial f}{\partial x} + \tfrac{1}{2}\sigma_t^2 \frac{\partial^2 f}{\partial x^2} \right) dt + \sigma_t \frac{\partial f}{\partial x} \, dW_t$$

Key Insight

The extra $\frac{1}{2}\sigma_t^2 f_{xx}$ term (absent in ordinary calculus) arises because Brownian increments satisfy $(\Delta W_t)^2 \approx \Delta t$, that is the quadratic variation of W is non-zero.

Black-Scholes PDE

Step 1 — Apply Itô's Lemma to $V(t, S_t)$:

$$dV_t = \left(\frac{\partial V}{\partial t} + \mu S \frac{\partial V}{\partial S} + \tfrac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} \right) dt + \sigma S \frac{\partial V}{\partial S}\, dW_t$$

Step 2 — Choose $\Delta_t = \partial V / \partial S$ to eliminate the dW_t risk:

$$d\Pi_t = \left(\frac{\partial V}{\partial t} + \tfrac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} \right) dt \quad \text{(locally risk-free)}$$

Step 3 — No-arbitrage requires $d\Pi_t = r\Pi_t\, dt$, giving:

Black-Scholes PDE

$$\frac{\partial V}{\partial t} + \tfrac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0, \qquad 0 \le t < T,\ S > 0$$

Boundary condition for a call: $V(T, S) = \max(S - K, 0)$. Closed-form solution: $C = S\,N(d_1) - Ke^{-r\tau} N(d_2)$, $\Delta = N(d_1)$.

Self-Financing Delta Hedge

Hold $\phi_t = \Delta_t \cdot n$ shares and a bond position (cash):

$$\text{cash}_0 = W_0 + C_0 \cdot n - \Delta_0 \cdot n \cdot S_0$$

Daily dynamics:

- Bond accrual: $\text{cash}_t \leftarrow \text{cash}_{t-1}\, e^{r/365}$
- Rebalance: $\text{cash}_t \leftarrow \text{cash}_t - (\phi_t - \phi_{t-1}) \cdot S_t$

Replication error and excess P&L:

$$\varepsilon_t = \underbrace{(\phi_t S_t + \text{cash}_t)}_{V_t^{\text{rep}}} - \underbrace{C_t \cdot n}_{L_t} \qquad \pi_t = \varepsilon_t - W_0 e^{r \cdot \text{elapsed}}$$

Contd.

Discrete Hedge Residual (Gamma-Theta P&L)

$$\Delta\pi_t \approx \underbrace{\tfrac{1}{2}\Gamma_t(\Delta S_t)^2}_{\text{gamma cost}} - \underbrace{\Theta_t \cdot \Delta t}_{\text{theta earned}}$$

Under *continuous* rebalancing $\pi_t = 0$ exactly; discretisation introduces residual noise.

Monte Carlo Delta Estimation

Under the risk-neutral measure, the terminal price follows

$$S_T = S_t \exp\big((r - \tfrac{1}{2}\sigma^2)\tau + \sigma\sqrt{\tau}\, Z\big), \quad Z \sim \mathcal{N}(0,1)$$

Pathwise MC estimator for Δ:

$$\hat{\Delta}^{\mathrm{MC}} = e^{-r\tau} \cdot \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}_{\{S_T^{(i)} > K\}} \cdot \frac{S_T^{(i)}}{S_t}$$

Variance reduction — antithetic variates: pair each draw Z with $-Z$, halving variance at no extra cost.

Contd.

MC Error Rate

$$\text{error} \sim O\left(\frac{1}{\sqrt{N}}\right)$$

Doubling path count $\Rightarrow \sqrt{2}\times$ accuracy improvement.

Higher-order Greeks (Γ, Θ, $\mathcal{V}$) fall back to `BlackCalculator` (closed-form) — MC second-order finite differences are too noisy at these path counts.

Backtesting Framework: Loom / Aurum

Loom

- Core C++ computation and streaming framework
- Provides derivative definitions, Black–Scholes models, and pricing/risk engines
- High-performance calculation DAG Builder

Aurum: Event-driven back-testing engine

- Represents market data, orders, fills, positions, and portfolio updates as events
- Supports consistent strategy simulation under evolving market states

Key guarantee:

Quiescing Invariant

Market updates, trade events, and portfolio events are serialized in a deterministic order, so strategy evaluation remains consistent across runs.

Prism: React visualisation dashboard

- Overview tab: replication error, excess P&L, hedge-quality table
- Detail tab: per-strategy playback with Greeks panel, portfolio card, event log

Experiment Setup

5 strategies on SPY data for over 6 months ($S_0 = 576.82$, $\sigma = 24.49\%$, $r = 4.5\%$).

Study A: Rebalancing Frequency (Analytic Δ)

Strategy	Interval	Trades
A_analytic_1d	Daily	113
A_analytic_5d	Weekly	29
A_analytic_20d	Monthly	8

Study B: MC vs Analytic Δ (weekly rebalance)

Strategy	Delta method	Paths
A_analytic_5d	Black-Scholes closed-form	—
B_mc_500	Pathwise MC (antithetic)	500
B_mc_5000	Pathwise MC (antithetic)	5000

Contd.

Initial position: $S_0 = \$576.82$, $\Delta_0 = 0.575$, $C_0 \cdot n = \$6{,}548$, $\phi_0 = 100$ shares, $\text{cash}_0 = \$0$

Study A: Effect of Rebalancing Frequency

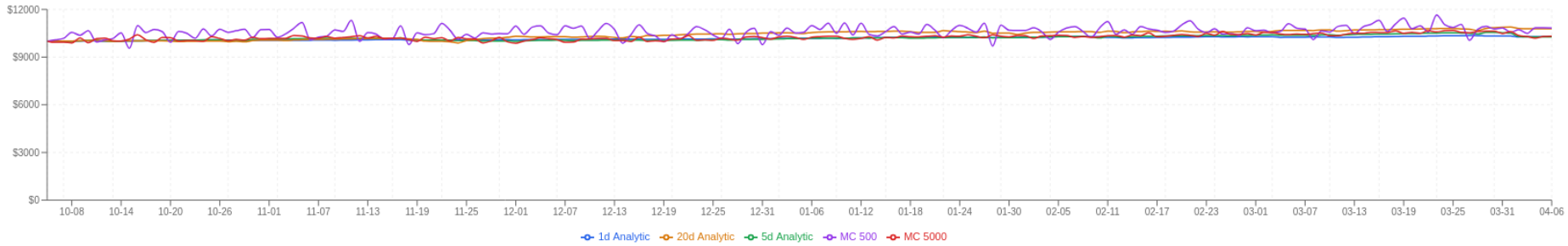

Figure 1: Replication Error ε_t

Key observations:

- *Daily* rebalancing tracks the option liability most closely — but incurs the most transaction cost (113 trades)
- *Monthly* rebalancing deviates most with only 8 trades
- Classic frequency–cost tradeoff

Gamma-Theta intuition:

- ε_t oscillates around benchmark
- Large $|\Delta S|$ days: gamma losses dominate
- Theta slowly earns back time value

Study A — Prism Detail View

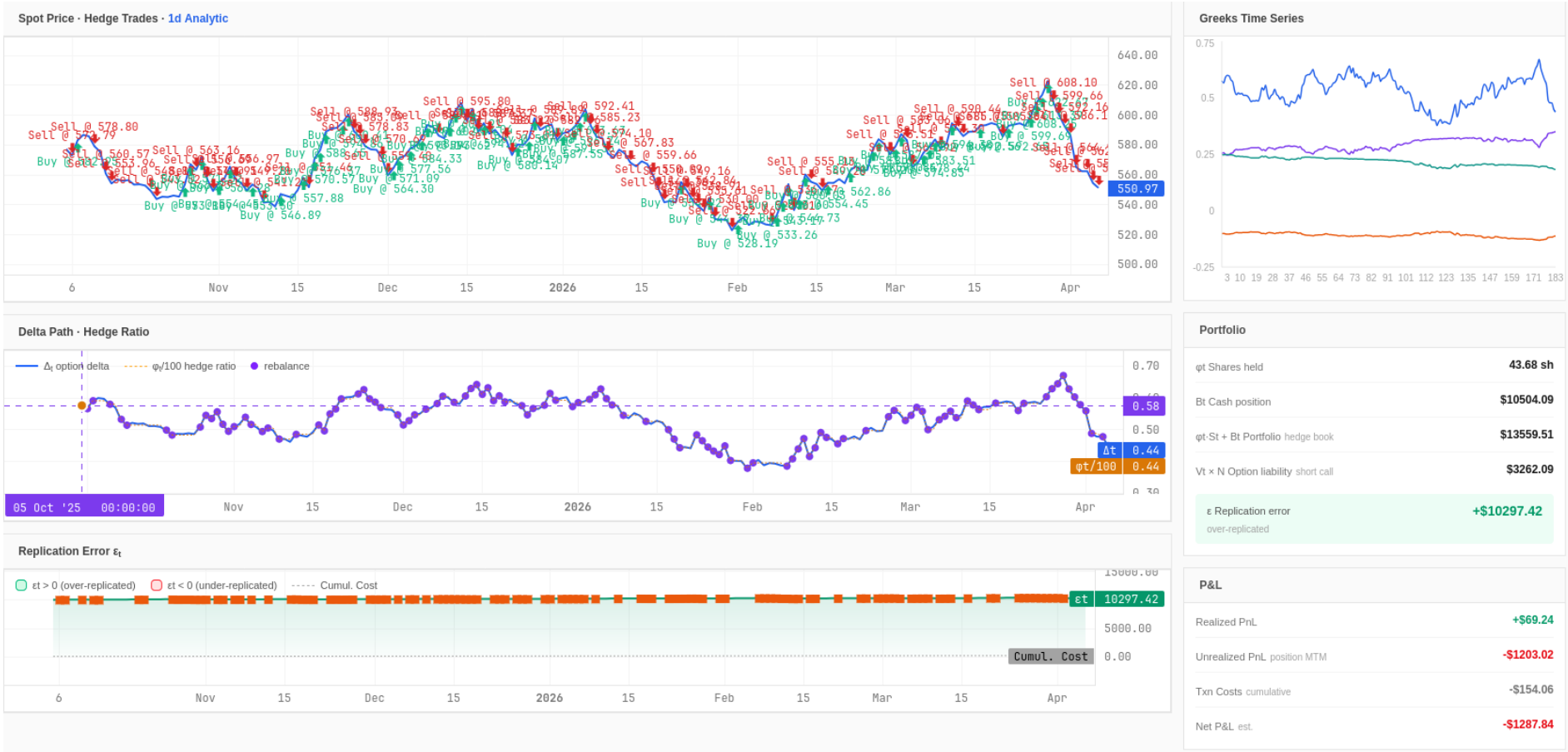

Figure 2: Daily Strategy Detail: Spot Path, BUY/SELL trade markers, Delta Path Over Time, Greeks Panel

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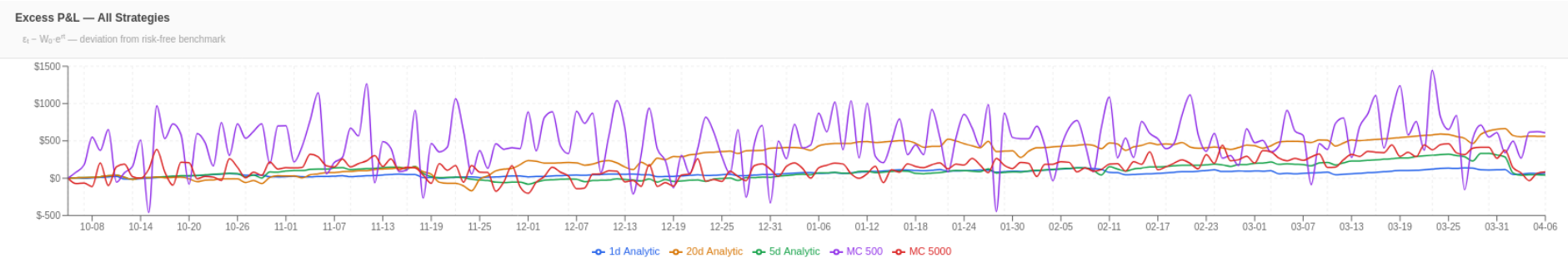


Figure 3: Excess P&L chart

Hedge Quality — Strategy Comparison

Strategy	Sharpe	RMSE	Max DD	Avg |ε|	Final ε	Excess P&L	Trades
1d Analytic	+0.53	$71.39	-$91.66	$10174.61	+$10297.42	+$69.24	113
5d Analytic	+0.13	$132.23	-$293.78	$10203.96	+$10270.34	+$42.16	29
20d Analytic	+1.28	$364.27	-$318.33	$10412.68	+$10787.27	+$559.09	8
MC 500	+0.11	$597.66	-$1709.74	$10610.18	+$10835.14	+$606.96	32
MC 5000	+0.06	$202.14	-$586.32	$10259.99	+$10313.35	+$85.17	30

Figure 4: Hedge Quality Table: Sharpe, RMSE, Max Drawdown

Study B — Monte Carlo Δ Convergence

Does MC delta converge to Black-Scholes?

Replication RMSE measures how much delta noise propagates into the hedge:

Strategy	RMSE	Excess P&L
5d Analytic	\$132.23	+\$42.16
MC 5000	\$202.14	+\$85.17
MC 500	\$597.66	+\$606.96

RMSE ratio: MC 500 / MC 5000 $= 597.66/202.14 \approx \mathbf{2.96}\times$
Theoretical $O(1/\sqrt{N})$ prediction: $\sqrt{5000/500} = \sqrt{10} \approx 3.16\times$

Key Result

MC 5000 converges toward the analytic hedge at close to the expected $O(1/\sqrt{N})$ rate ($2.96\times$ vs $3.16\times$ predicted).

Study B — MC vs Analytic: Replication Error

How does delta noise propagate into the hedge?

All three strategies rebalance on a *weekly* schedule on the same dates. The only difference is $\hat{\Delta}$.

Terminal ε_T (from Prism)

Strategy	Final ε_T	Gap from analytic
5d Analytic	+\$10,270.34	—
MC 5000	+\$10,313.35	+\$43
MC 500	+\$10,835.14	+\$565

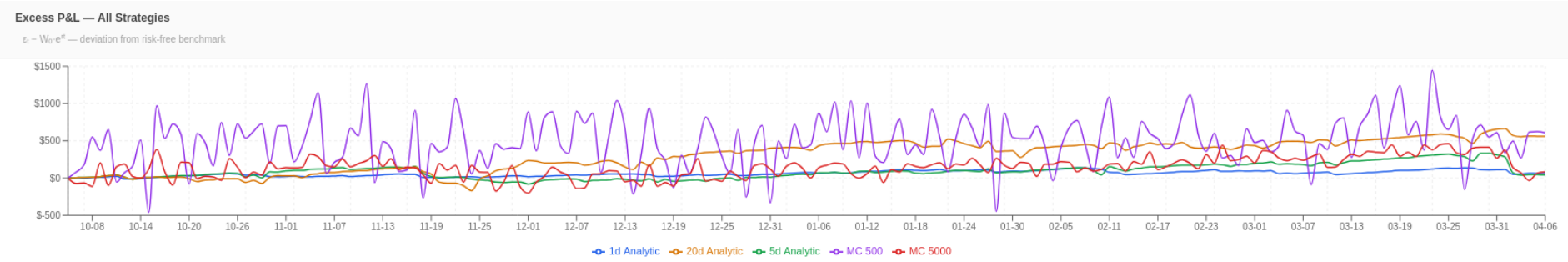


Figure 5: Excess P&L chart

Conclusions

Study A — Rebalancing Frequency

- Higher rebalancing frequency reduces replication error: RMSE \$71 (daily) $<$ \$132 (weekly) $<$ \$364 (monthly)
- Daily used 113 trades vs. 8 for monthly — a 14$\times$ difference in transaction cost exposure

Study B — Monte Carlo Convergence

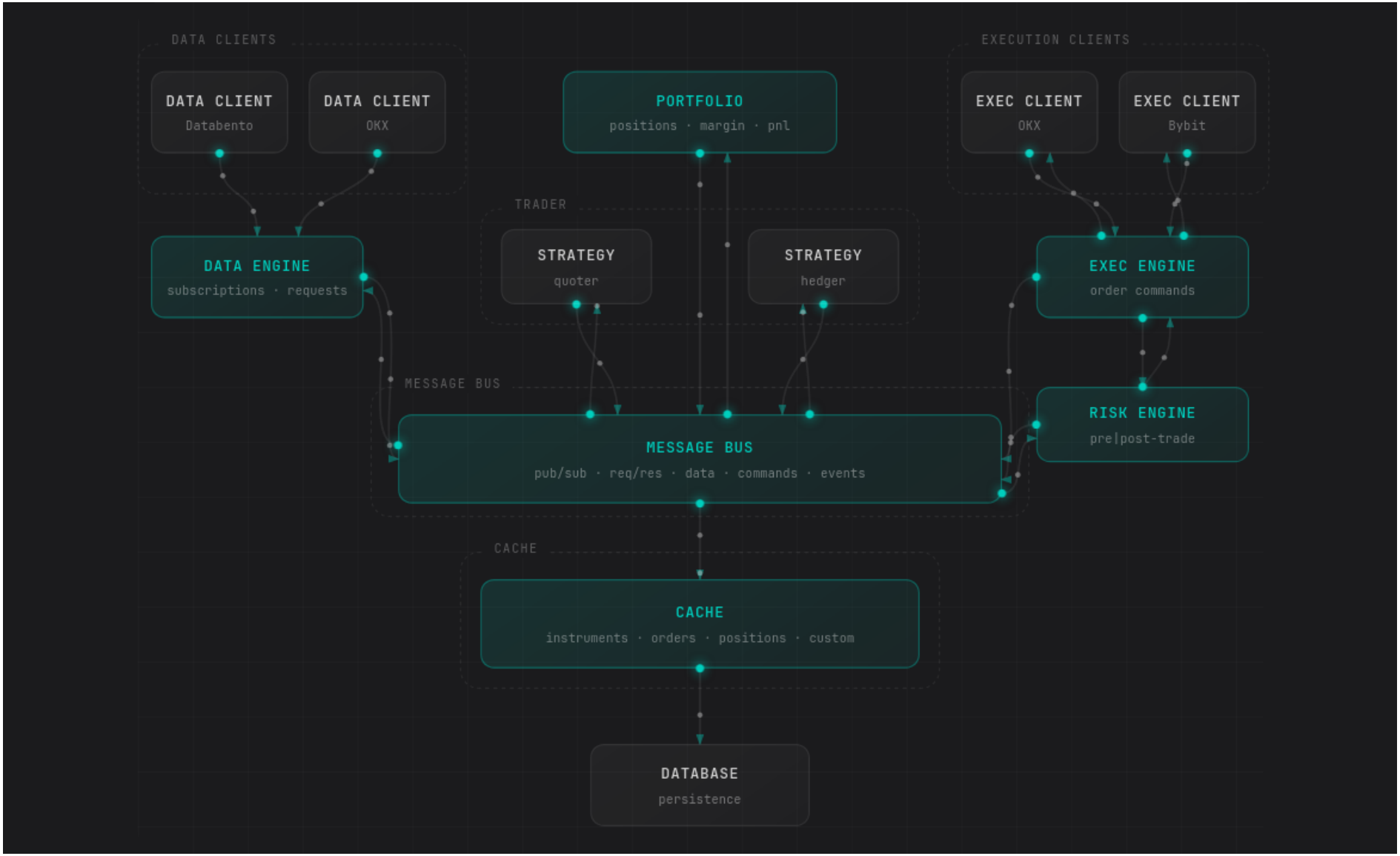
- RMSE ratio MC 500 / MC 5000 $\approx 2.96\times$, close to the theoretical $\sqrt{10} \approx 3.16\times$ prediction
- MC 5000 terminal ε_T is only \$43 from the analytic result; MC 500 is \$565 away ($\approx$13$\times$ the gap)

Contd.

Take-Away

Discrete delta hedging succeeds in replicating the option payoff with manageable error. The Gamma-Theta balance governs hedge quality; more frequent rebalancing reduces tracking error at the cost of higher transaction volume.

Group Contributions

Member	Contributions
Quin	Full C++ engine implementation (Loom streaming library, Aurum trading engine, `BlackCalculator`, delta-hedge strategies, GBM datafeed, CSV output)
Prabhat	Mathematical framework documentation (Itô's Lemma derivation, BS PDE, self-financing portfolio analysis), results analysis, Prism visualisation, presentation preparation
Andrew	Monte Carlo option pricing theory, variance reduction analysis, Study B experimental design, Experiment setup, data analysis

Engine Structure